Varendra University

Assignment for 3rd Year 1st Semester

Department of Computer Science and Engineering

Course: CSE 3107 (Communication Engineering) **Total Marks:** 15

Title: "Design an Optical Fiber Communication Network for Smart Varendra University Campus"

Scenario

The admin of Varendra University plans to provide reliable high-speed Internet connectivity throughout the campus. As a communication engineer, you are assigned to design an optical fiber communication system for this purpose.

You must address the following task:

1. **Application Scenario**
   - Justify why optical fiber is preferred over conventional media on your campus.
2. **System Architecture**
   - Draw a generic functional block diagram of the fiber optic communication system to provide the connection in the CSE classrooms.
   - Explain the role of each component.
3. **Fiber Selection**
   - Choose between single-mode and multimode fiber.
   - Justify the choice based on distance and bandwidth requirements.
4. **Fiber Structure**
   - Describe the core, cladding, and coating and explain their significance.
5. **Propagation Mechanism**
   - Explain how light propagates inside the fiber considering Reflection and Refraction, Refractive index, Critical angle, Total internal reflection
6. **Loss Analysis**
   - Identify possible intrinsic and extrinsic losses that may occur in your proposed network and suggest how to minimize these losses.
7. **Performance Considerations**
   - Discuss the ideal characteristics expected from the communication system.

Students must submit:

1. A **4–6 page handwritten report on June 15, 2026**
2. Include diagrams and figures wherever appropriate